AI-Powered Snake Species Identification for Antivenom Selection

Engineering a deep learning pipeline to solve the 60-minute survival window.

PRESENTED BY

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The Critical Bottleneck in Emergency Healthcare

The Scale

58,000

Deaths Per Year in India alone.
Snakebites claim more lives in this region than any other natural emergency.

The Time Constraint

The 'Golden Hour' Criterion:

Antivenom selection within one hour dramatically improves survival rates. Manual species identification by remote healthcare workers is currently unreliable and far too slow.

Engineering a Clinical-Grade Solution

01. High-Precision Classification

Develop deep learning models capable of classifying snake species directly from raw, field-captured images.

02. Maximising Recall

Ensure zero misses for venomous species; treating false negatives as a life-threatening failure condition.

03. Model Evolution

Systematically compare baseline algorithms, custom CNNs, and transfer learning to identify the optimal deployment architecture.

04. Clinical Explainability

Provide visual, medical-grade validation using Grad-CAM to prove the model's reasoning to healthcare professionals.

The Data Funnel: Distilling the Kaggle Snake Species Dataset

RAW DATASET

23,816

training images

135 species · 6 families

36.5% venomous · 63.5% non

512 × 512 px originals

FILTER

Top 10

most-represented species

selected to build a clean, class-balanced training set

WORKING SUBSET

2,006

curated images

a focused, high-quality base ready for preprocessing

WHY DISTILL? Most of the 135 species have too few images to learn from, so we focus on the 10 best-represented — a clean subset of 2,006 images that trains reliably under limited compute.

CLASS BALANCE 5 venomous / 5 non-venomous species (~50 / 50 split).

Preprocessing & Augmentation: Preparing Images for Training

1
Resize
Every image is scaled to 224 × 224 px (RGB) — the native input size of the CNN backbone.

2
Normalise
Pixel values are rescaled from 0–255 down to the [0, 1] range so the network trains stably.

3
Stratified Split
An 80 / 20 train–validation split preserves each species' proportion across both sets.

4
Augment
Training set only: random flip, rotation and zoom add variety and curb overfitting.

224 × 224
INPUT SIZE · RGB

1,604
TRAINING IMAGES · 80%

402
VALIDATION IMAGES · 20%

Architectural Evolution: From Baseline to State-of-the-Art

Baseline (Random Forest)

Approach

Classic Machine Learning

Mechanism

GridSearchCV optimisation

Features

150K+ flattened features

Model 1 (Custom CNN)

Approach

Deep Learning from Scratch

Mechanism

3 custom-trained conv blocks

Features

Acts as a baseline for feature extraction

Model 2 (Transfer Learning)

Approach

State-of-the-art Feature Extraction

Mechanism

MobileNetV2 backbone (frozen)

Features

Paired with a custom classification head

Baseline Report: Random Forest + GridSearchCV

Species	Precision	Recall	F1-score	Support
⚠️ *Agkistrodon piscivorus*	0.17	0.17	0.17	40
Ahaetulla nasuta	0.23	0.12	0.16	40
Ahaetulla prasina	0.35	0.17	0.23	40
⚠️ *Austrelaps superbus*	0.22	0.25	0.24	40
Boiga kraepelini	0.35	0.33	0.34	40
⚠️ *Bothriechis schlegelii*	0.27	0.31	0.29	42
⚠️ *Bothrops asper*	0.17	0.10	0.12	41
Carphophis vermis	0.27	0.49	0.35	39
⚠️ *Crotalus tigris*	0.32	0.42	0.37	40
Imantodes cenchoa	0.30	0.33	0.31	40
Macro avg	**0.27**	**0.27**	**0.26**	**402**
Weighted avg	**0.27**	**0.27**	**0.26**	**402**

VALIDATION ACCURACY

26.9%

402 validation images · 10 classes

THE TAKEAWAY

A classical Random Forest tops out at ~27% — well above the 10% random-chance floor for 10 classes, but far short of clinical reliability.

This gap is exactly what deep transfer learning is built to close.

CNN Report: Custom Convolutional Network

Species	Precision	Recall	F1-score	Support
⚠️ *Agkistrodon piscivorus*	0.28	0.33	0.30	40
Ahaetulla nasuta	0.55	0.40	0.46	40
Ahaetulla prasina	0.47	0.45	0.46	40
⚠️ *Austrelaps superbus*	0.39	0.45	0.42	40
Boiga kraepelini	0.43	0.38	0.40	40
⚠️ *Bothriechis schlegelii*	0.56	0.64	0.60	42
⚠️ *Bothrops asper*	0.39	0.34	0.36	41
Carphophis vermis	0.76	0.67	0.71	39
⚠️ *Crotalus tigris*	0.45	0.70	0.55	40
Imantodes cenchoa	0.46	0.33	0.38	40
Macro avg	**0.48**	**0.47**	**0.47**	**402**
Weighted avg	**0.48**	**0.47**	**0.47**	**402**

VALIDATION ACCURACY

46.8%

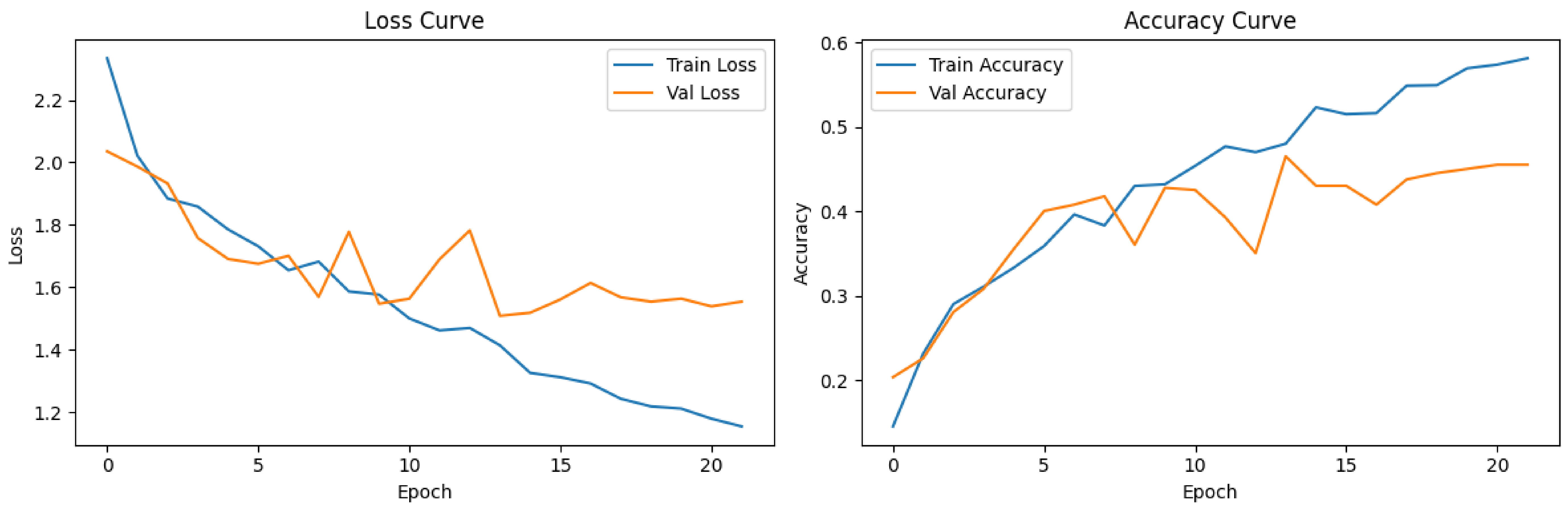
402 validation images · 10 classes

THE TAKEAWAY

A purpose-built CNN nearly doubles the baseline — 26.9% → 46.8% (+19.9 pts). Learned visual features clearly help, yet accuracy still falls short of clinical reliability.

Transfer learning takes the next leap.

CNN Training Dynamics



Train accuracy keeps climbing while validation accuracy plateaus near 47% and validation loss begins to rise after ~epoch 13 — classic signs of overfitting in the from-scratch CNN, motivating the shift to transfer learning.

Transfer Learning Report: MobileNetV2 (Champion)

Species	Precision	Recall	F1-score	Support
⚠️ *Agkistrodon piscivorus*	0.64	0.57	0.61	40
Ahaetulla nasuta	0.58	0.65	0.61	40
Ahaetulla prasina	0.56	0.47	0.51	40
⚠️ *Austrelaps superbus*	0.73	0.88	0.80	40
Boiga kraepelini	0.56	0.70	0.62	40
⚠️ *Bothriechis schlegelii*	0.77	0.71	0.74	42
⚠️ *Bothrops asper*	0.84	0.63	0.72	41
Carphophis vermis	0.90	0.97	0.94	39
⚠️ *Crotalus tigris*	0.87	0.97	0.92	40
Imantodes cenchoa	0.69	0.55	0.61	40
Macro avg	**0.71**	**0.71**	**0.71**	**402**
Weighted avg	**0.71**	**0.71**	**0.71**	**402**

VALIDATION ACCURACY

71.1%

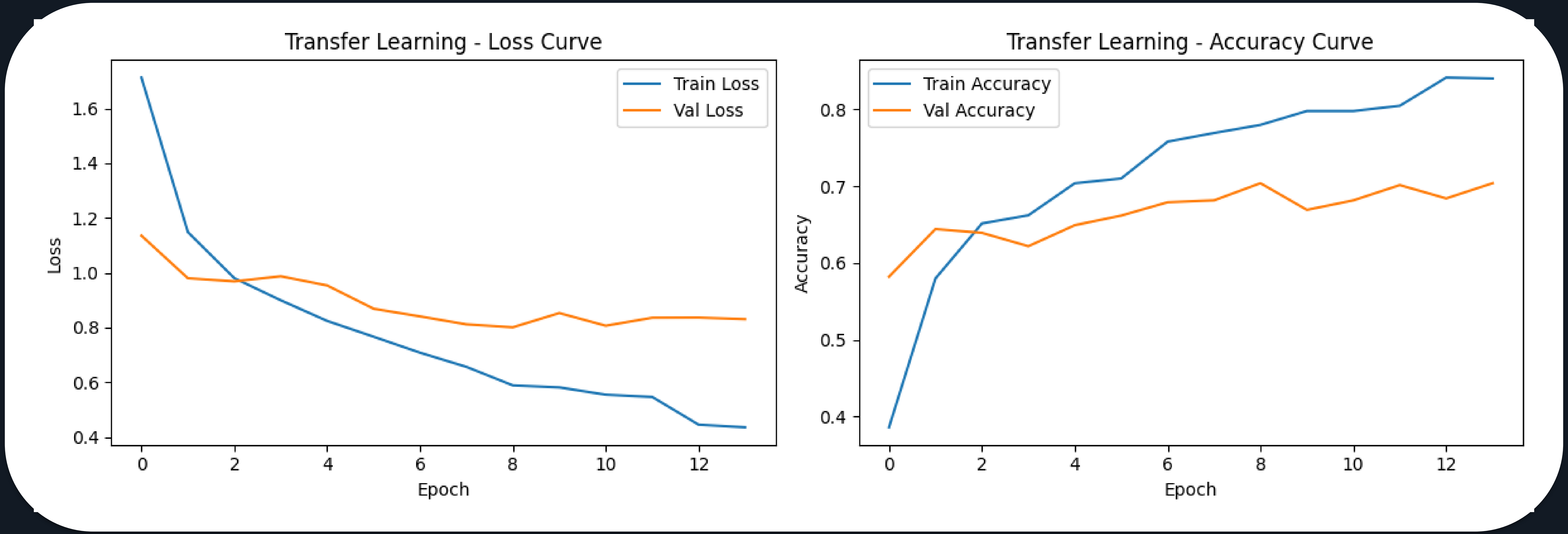
402 validation images · 10 classes

THE TAKEAWAY

Transfer learning delivers the leap — 46.8% → 71.1% (+24.4 pts over the from-scratch CNN, +44.3 pts over baseline). Pre-trained ImageNet features generalise far better on a small dataset.

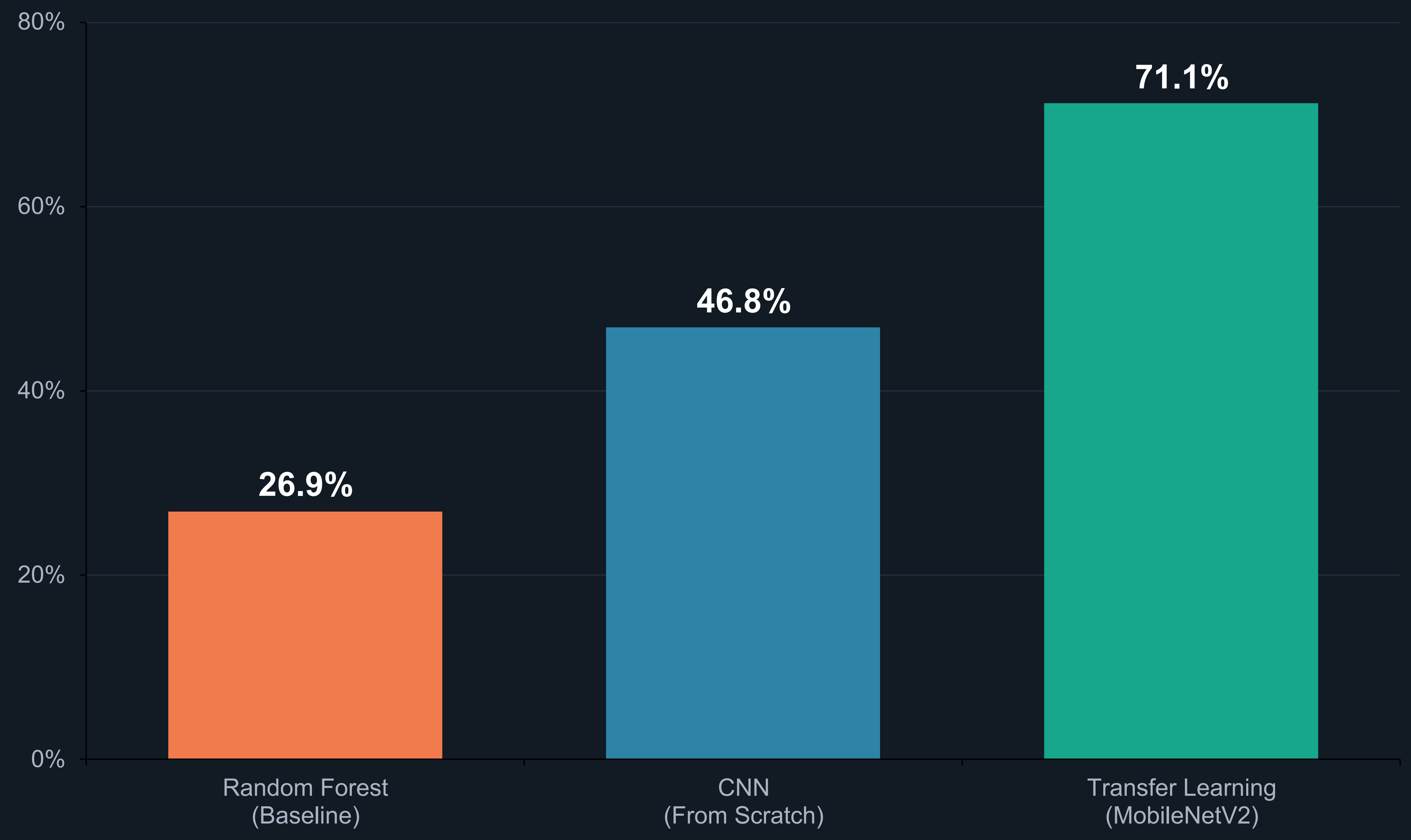
This is the champion model.

Transfer Learning Dynamics



Validation accuracy climbs steadily to ~70% while validation loss stays low and flat — the pretrained ImageNet features generalise well, with only a modest train/validation gap and no runaway overfitting.

Performance Benchmark:
The Transfer Learning Advantage



A **+44.3 pt** absolute gain over the baseline.

Pre-trained ImageNet weights successfully capture the complex, nuanced visual patterns of snake scales that custom models struggle to learn from scratch.

The Champion Architecture: Efficient by Design

Input
224 × 224 × 3

→

MobileNetV2
frozen backbone · 7×7×1280

→

Global Avg Pool
→ 1280

→

Dense 128
ReLU + Dropout

→

Dense 10
Softmax → 10 classes

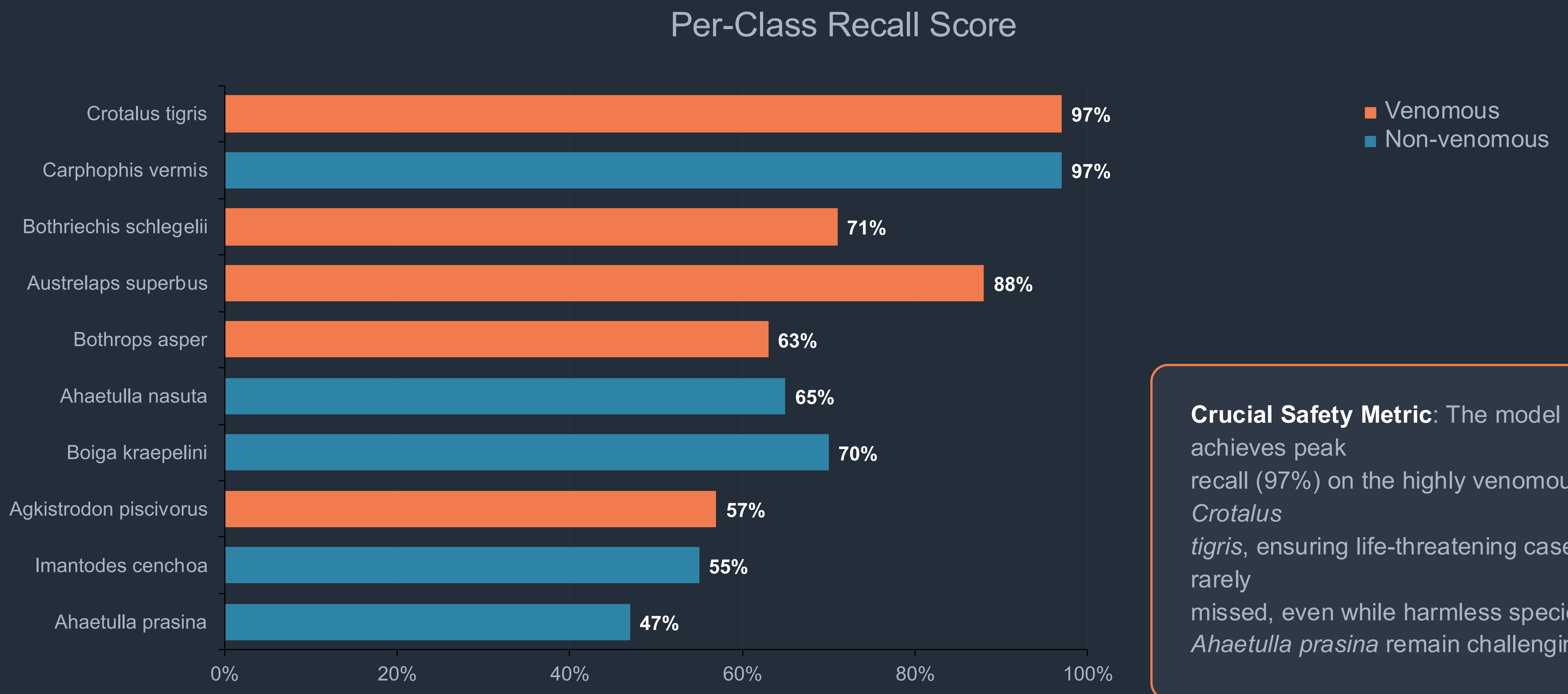
MODEL SIZE
9.24 MB
2,423,242 total parameters

TRAINABLE
165 K
just 6.8% — the classifier head

FROZEN BACKBONE
8.61 MB
2.26M pretrained ImageNet params

WHY IT'S THE CHAMPION Freezing the pretrained backbone and training only a 165K-parameter head keeps the model ~9 MB and quick to train — accurate yet light enough for on-device, offline deployment in the field.

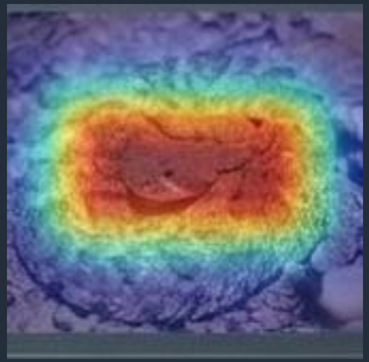
Safety First: Per-Class Recall and Venomous Profiling



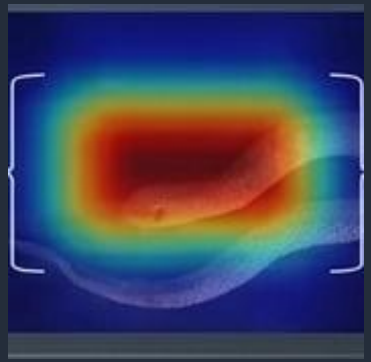
Crucial Safety Metric: The model achieves peak recall (97%) on the highly venomous *Crotalus tigris*, ensuring life-threatening cases are rarely missed, even while harmless species like *Ahaetulla prasina* remain challenging.

Clinical Explainability: Unlocking the 'Black Box' with Grad-CAM

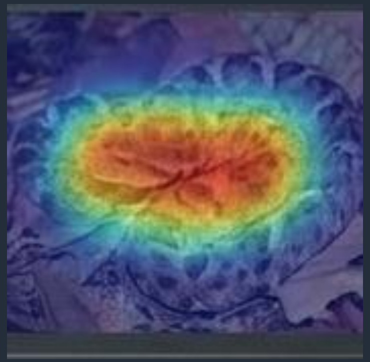
Crotalus tigris (93%)



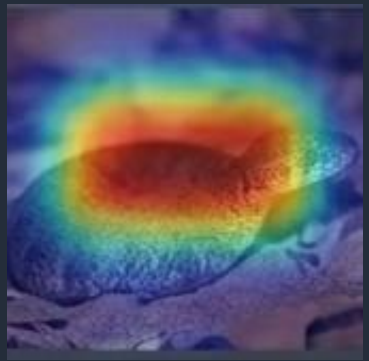
Carphophis vermis (92%)



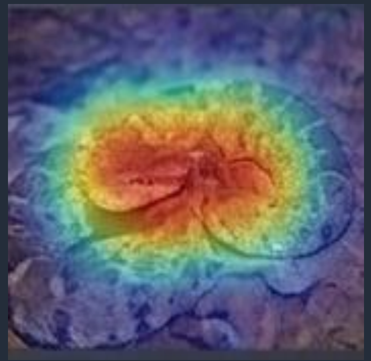
Bothrops asper (76%)



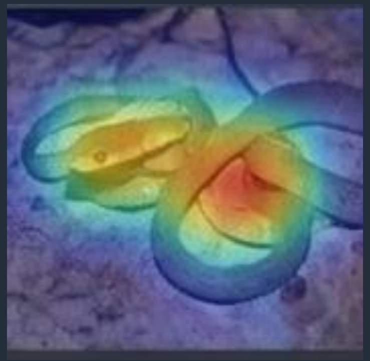
Austrelaps superbus (82%)



Bothriechis schlegelii (81%)



Boiga kraepelini (68%)



Attention maps confirm the model's clinical viability. Predictions are driven by relevant anatomical features (cranial shapes, specific scale patterns, body morphology) rather than spurious background artifacts like leaves or rocks.

Real-World Impact: From Algorithm to Emergency Response

Emergency Response

Reduces species identification in rural clinics from hours of manual searching to mere minutes.

Life-Saving Decisions

Enables immediate, accurate antivenom selection, dramatically improving the odds of survival within the Golden Hour.

Scalable Reach

Optimised for mobile apps, telemedicine platforms, and offline field-based wildlife management tools in remote regions.

Current Limitations and Clinical Boundaries

Subset Dependency

The model was trained on 2,006 images (top 10 species) out of the full 23,816. Expanding to all 135 species requires exponentially more compute.

Image Quality

Field photos suffer from poor lighting, motion blur, and occlusion. Real-world robustness is highly dependent on input quality.

Class Imbalance

Certain morphological overlaps cause severe misclassification risks for specific species (e.g., Ahaetulla prasina).

Pre-Clinical Status

The algorithm requires rigorous, formal testing with medical professionals and toxicologists before live clinical deployment.

Key Takeaways

Accuracy (71.1%)

Transfer learning decisively turns a notoriously difficult 135-species problem into a tractable mathematical classifier.

Venomous Recall (97%)

The system inherently protects patients by heavily indexing on the highest-risk species like *Crotalus* tigris.

Grad-CAM Validation

Anatomical feature mapping proves the model acts as an explainable medical assistant, not a blind algorithm.

Conclusion: By pairing MobileNetV2 with explainable AI, we have engineered a system that is effective, transparent, and structurally sound. It provides a definitive foundation for clinical tools capable of saving thousands of lives.

Future Work & Roadmap

1. **Mobile & Edge Deployment**
   Convert to TensorFlow Lite for fast, offline on-device inference in disconnected rural settings.

2. **Fine-Tune & Ensemble Models**
   Unfreeze MobileNetV2's top layers and explore ViT / EfficientNet ensembles for further accuracy.

3. **Broader, Region-Specific Data**
   Scale to 10,000+ images across 50+ species and train region-specific models for local accuracy.

4. **Multi-Modal & GPS Context**
   Add audio, thermal and location signals to narrow likely species and raise confidence.

5. **Antivenom Inventory Integration**
   Link predictions to hospital antivenom stock for end-to-end emergency response.

References

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Questions & Discussion

Thank you for your attention.